Universal Resonance Calculus — Formal Core

1) Universe and Signature

- **Base set**: $\mathbb{N}_0 = \{0, 1, 2, \dots\}$.
- **Lifted set**: $D = \mathbb{N}_0 \cup \{\perp\}$.
- **Operations on** D :
- Addition $\oplus : D \times D \to D$
- Multiplication $\otimes : D \times D \to D$
- Distinguished element $\perp$ (error/undefined sentinel)

Axioms 1. $(\mathbb{N}_0, +, \cdot, 0, 1)$ is the standard commutative semiring. 2. For $x, y \in D$: if $x = \perp$ or $y = \perp$ then $x \oplus y = \perp$ and $x \otimes y = \perp$. Otherwise $x \oplus y = x + y$ and $x \otimes y = xy$. 3. $\perp$ is absorbing for both operations and not equal to any natural number.

> This keeps the standard arithmetic on $\mathbb{N}_0$ while offering a total, conservative extension via $\perp$.

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2) Binary Encoding and Weights

- For $n \in \mathbb{N}_0$ let $b_i(n) \in \{0, 1\}$ be the i -th bit of its binary expansion: $n = \sum_{i \ge 0} b_i(n)\, 2^i$.
- Fix a **strictly positive absolutely summable** weight sequence $w \in \ell^1(\mathbb{N}_0)$ with $w_i > 0$ and $\sum_{i \ge 0} w_i = W < \infty$.
- Choose a parameter vector $\alpha = (\alpha_i)_{i \ge 0}$ with $\alpha_i \in [\varepsilon, 1 - \varepsilon]$ for some fixed \varepsilon\in(0,\tfrac12]\.

Canonical choice: $w_i = 2^{-(i+1)}$ so $W = 1$.

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3) Resonance Map

Define the **resonance** of n as

$$r_\alpha(n) \;=\; \sum_{i=0}^{\infty} \alpha_i\, b_i(n)\, w_i \;\in [0, W].$$

Properties 1. **Boundedness**: $0 \le r_\alpha(n) \le W$ for all n . 2. **Lipschitz in parameters**: for any α, β , $|r_\alpha(n) - r_\beta(n)| \le \|\alpha - \beta\|_\infty W$. 3. **Tail error under truncation**: with the truncation level K , let $r_\alpha^{(K)}(n) = \sum_{i=0}^{K-1} \alpha_i b_i(n) w_i$. Then for all n

$$|r_\alpha(n) - r_\alpha^{(K)}(n)| \le \sum_{i \ge K} w_i.$$

4. **Aperiodicity (generic)**: if the set $\{\alpha_i w_i\}_{i\ge 0}$ is linearly independent over $\mathbb{Q}$ and infinitely many $w_i > 0$, then the map $n \mapsto r_\alpha(n)$ is aperiodic.

> Use the infinite support of w to avoid finite-bit periodicity. Finite computations use K with a certified tail bound.

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4) Discrete Calculus on r_α

Define forward difference and discrete Hessian:

$$\Delta r_\alpha(n) = r_\alpha(n{+}1) - r_\alpha(n), \quad \Delta^2 r_\alpha(n) = \Delta r_\alpha(n{+}1) - \Delta r_\alpha(n).$$

Local minima (strict): n is a strict local minimum of r_α if $\Delta r_\alpha(n{-}1) < 0$ and $\Delta r_\alpha(n) > 0$. Ties are broken by choosing the smallest index in any flat run.

All definitions apply verbatim to $r_\alpha^{(K)}$.

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5) Page Structure (Finite Windows)

Fix $K \ge 1$ and set the **page base** $B = 2^K$. For $n \in \mathbb{N}_0$:

$$\mathrm{page}(n) = \lfloor n/B \rfloor, \qquad \mathrm{offset}(n) = n \bmod B.$$

Analyses and training below operate per page or on a finite union of pages.

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6) Finite-Window Learning Task

Let a finite index set $W \subset \mathbb{N}_0$ be given. Optionally let labels $y : W \to \{0,1\}$ be given (e.g., any binary property of integers). Define the feature map

$$\varphi_\alpha^{(K)}(n) \;=\; \big(r_\alpha^{(K)}(n),\; \Delta r_\alpha^{(K)}(n),\; \Delta^2 r_\alpha^{(K)}(n)\big) \in \mathbb{R}^3.$$

Pick a smooth loss $\ell : \mathbb{R}^3 \times \{0,1\} \to \mathbb{R}_{\ge 0}$ and set the empirical loss

$$L_{W,K}(\alpha) \;=\; \frac{1}{|W|} \sum_{n\in W} \ell\big(\varphi_\alpha^{(K)}(n),\, y(n)\big).$$

Projection set: $A_K = [\varepsilon, 1{-}\varepsilon]^K$ (compact, convex in $\mathbb{R}^K$).

Update map (fixed step $\eta > 0$):

$$F(\alpha) \;=\; \Pi_{A_K}\big(\alpha - \eta\, \nabla L_{W,K}(\alpha)\big), \quad F : A_K \to A_K.$$

If ℓ is C^1 , then F is continuous. By **Brouwer**, F admits a fixed point $\alpha^\star \in A_K$, i.e., $F(\alpha^\star) = \alpha^\star$.

> $\alpha^\star$ is a stationary point of the projected learning dynamics on the finite-dimensional parameter space.

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7) Guarantees and Bounds

- **Operator-norm bound**: define the multiplication operator $(M_{r_\alpha} f)(n) = r_\alpha(n) f(n)$ on $\ell^2(\mathbb{N}_0)$. Then $\|M_{r_\alpha}\| = \sup_n |r_\alpha(n)| \le W$.
- **Shift-coupled operator**: with the shift $(Sf)(n) = f(n{+}1)$ and any $\rho \in \mathbb{R}$, set $T_{\alpha,\rho} = S - \rho M_{r_\alpha}$. Then $\|T_{\alpha,\rho}\| \le 1 + |\rho|\, W$.
- **Truncation error**: replacing r_α by $r_\alpha^{(K)}$ perturbs all values and finite-difference features by at most $2 \sum_{i \ge K} w_i$ pointwise.
- **Aperiodicity in practice**: choosing α_i i.i.d. from a continuous distribution on $[\varepsilon, 1{-}\varepsilon]$ yields aperiodicity with probability 1.

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8) Algorithms (Reference)

A) Encoding and Resonance (truncated) 1. Input: $n, K, \alpha \in A_K, w$. 2. Compute bits $b_0(n), \dots, b_{K-1}(n)$. 3. Return $r_\alpha^{(K)}(n) = \sum_{i=0}^{K-1} \alpha_i b_i(n) w_i$.

B) Minima Detection on a Window 1. Input: window $W = \{m, \dots, m+L\}$. 2. Compute $r, \Delta r, \Delta^2 r$ for all indices in W . 3. Output strict local minima $\{n \in W : \Delta r(n{-}1) < 0,\ \Delta r(n) > 0\}$ with tie-breaking by smallest index.

C) Projected Gradient Step 1. Input: $\alpha \in A_K$, labels y on W , loss ℓ , step η . 2. Compute $\nabla L_{W,K}(\alpha)$ via autodiff or analytic gradients. 3. Update $\alpha \leftarrow \Pi_{A_K}(\alpha - \eta\, \nabla L_{W,K}(\alpha))$.

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9) Optional Spectral View

Define $T_{\alpha,\rho} = S - \rho M_{r_\alpha}$ on ℓ^2 . Spectral radii $\rho(T_{\alpha,\rho})$ control linear stability of the coupled shift–resonance dynamics. With $\|M_{r_\alpha}\| \le W$, any small enough $|\rho|$ ensures $\|T_{\alpha,\rho}\| < 1 + \delta$ for prescribed $\delta > 0$. This gives a tunable stability certificate for algorithms driven by r_α .

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10) Parameter Choices and Defaults

- Weights: $w_i = 2^{-(i+1)}$.
- Truncation: choose K so that $\sum_{i \ge K} w_i \le \tau$ for tolerance τ .
- Parameter box: $A_K = [\varepsilon, 1{-}\varepsilon]^K$ with $\varepsilon = 10^{-3}$ unless otherwise required.
- Loss example: logistic or squared loss on the features $\varphi_\alpha^{(K)}(n)$.

11) Notation Summary

- $\mathbb{N}_0$: nonnegative integers. $D = \mathbb{N}_0 \cup \{\perp\}$.
- $b_i(n)$: binary bits. $w \in \ell^1$: positive weights, sum W .
- α : parameter vector; A_K : compact convex parameter set.
- r_α , $r_\alpha^{(K)}$: resonance (full and truncated).
- Δ, Δ^2 : discrete gradient and Hessian.
- **page**, **offset** with base $B = 2^K$.
- $L_{W,K}$: empirical loss; F : projected gradient map; $\alpha^\star$: Brouwer fixed point.

Minimal Example (Concrete Numbers)

Let $w_i = 2^{-(i+1)}$, $K = 12$, $\varepsilon = 10^{-3}$. For a single page take $W = \{0, 1, \ldots, 2^{12} - 1\}$. Initialize $\alpha_i = \frac{1}{2}$. Compute $r_\alpha^{(12)}$, its differences, and the minima set. If labels y are supplied on W , run the projected gradient map F until a stationary point is reached. Tail error is $\leq 2^{-12}$.